# MAT2002 HW1 文杰 123090612

## 1

Second order; nonlinear

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## 2

Given

$$y(t) = e^{t^2}\int_0^t e^{-s^2}ds + e^{t^2} = e^{t^2}\Big(\int_0^t e^{-s^2}ds + 1\Big),$$

let $I(t) = \int_0^t e^{-s^2}ds$. Then $I'(t) = e^{-t^2}$. Additionally, $\frac{d}{dt}e^{t^2} = 2te^{t^2}$. Hence,

$$\begin{aligned} y'(t) &= \frac{d}{dt}\Big[e^{t^2}(I(t)+1)\Big] \\ &= (2te^{t^2})(I(t)+1) + e^{t^2}\cdot I'(t) \\ &= 2te^{t^2}(I(t)+1) + e^{t^2}e^{-t^2} \\ &= 2ty(t) + 1. \end{aligned}$$

Therefore,

$$y'(t) - 2ty(t) = (2ty(t)+1) - 2ty(t) = 1$$

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## 3

The ODE is rewritten in the standard linear form $y' = p(t)y + q(t)$:

$$y' - 2y = e^{2t} \quad\Longleftrightarrow\quad y' = 2y + e^{2t},$$

so $p(t) = 2$ and $q(t) = e^{2t}$. Since $p, q$ are continuous for all $t \in \mathbb{R}$, the IVP has a **unique** solution defined on all $\mathbb{R}$.

The integrating factor is

$$\mu(t) = e^{-\int 2dt} = e^{-2t}.$$

Multiply the equation $y' = 2y + e^{2t}$ by $\mu(t)$:

$$e^{-2t}y' - 2e^{-2t}y = e^{-2t}e^{2t} = 1.$$

The left-hand side is $\dfrac{d}{dt}\big(e^{-2t}y\big)$, hence

$$\frac{d}{dt}\big(e^{-2t}y\big) = 1.$$

Integrating yields:

$$e^{-2t}y = t + C \quad\Longrightarrow\quad y(t) = e^{2t}(t + C).$$

Applying $y(0) = 2$:

$$2 = e^{0}(0 + C) = C \Rightarrow C = 2.$$

$$\therefore y(t) = e^{2t}(t+2)$$

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## 4

### (a) Homogeneous case $g(t) \equiv 0$

Then

$$y' + p(t)y = 0 \quad\Longleftrightarrow\quad y' = -p(t)y.$$

Assuming $y \neq 0$ on an interval, divide by $y$:

$$\frac{y'}{y} = -p(t) \quad\Longleftrightarrow\quad \frac{d}{dt}\ln|y(t)| = -p(t).$$

Integrate:

$$\ln|y(t)| = -\int p(t)dt + C \quad\Longleftrightarrow\quad y(t) = A\exp\left[-\int p(t)dt\right],$$

where $A = \pm e^C$ is a constant.

## (b) Variation of parameters ansatz

Assume

$$y(t) = A(t)\exp\left[-\int p(t)dt\right]. \tag{1}$$

Differentiate using product + chain rules:

$$\begin{aligned} y' &= A'(t)e^{-\int p} + A(t)e^{-\int p}\cdot\frac{d}{dt}\left(-\int p(t)dt\right) \\ &= A'(t)e^{-\int p} - p(t)A(t)e^{-\int p}. \end{aligned}$$

Substitute into $y' + p(t)y = g(t)$:

$$\left(A'e^{-\int p} - pAe^{-\int p}\right) + p\left(Ae^{-\int p}\right) = A'(t)e^{-\int p(t)dt} = g(t).$$

Hence

$$A'(t) = g(t)\exp\left[\int p(t)dt\right], \tag{2}$$

as required.

## (c) Solve for $A(t)$, then $y(t)$; verify agreement with integrating factor formula

From (2),

$$A(t) = \int g(t)\exp\left[\int p(t)dt\right]dt + C,$$

so by (1),

$$\boxed{y(t) = \exp\left[-\int p(t)dt\right]\left(\int g(t)\exp\left[\int p(t)dt\right]dt + C\right).} \tag{3}$$

Now compare with the integrating factor form. For

$$y' + p(t)y = g(t),$$

the integrating factor is

$$\mu(t) = \exp\left(\int p(t)dt\right),$$

so multiplying the ODE by $\mu$ gives $\frac{d}{dt}(\mu y) = \mu g$, hence

$$y(t) = \frac{1}{\mu(t)}\left(\int \mu(t)g(t)dt + C\right),$$

which is exactly (3) after substituting $\mu(t) = e^{\int p}$.

Finally, using a definite-integral version with base point $t_0$, define

$$\mu(t) := \exp\left(\int_{t_0}^{t} p(s)ds\right) \quad \Rightarrow \quad \mu(t_0) = 1.$$

Integrating $\frac{d}{dt}(\mu y) = \mu g$ from $t_0$ to $t$ yields

$$\mu(t)y(t) - \mu(t_0)y(t_0) = \int_{t_0}^{t}\mu(s)g(s)ds$$

so

$$\boxed{y(t) = \frac{1}{\mu(t)}\left[\int_{t_0}^{t}\mu(s)g(s)ds + c\right],}$$

(where $c = \mu(t_0)y(t_0) = y(t_0)$ for an IVP).

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# 5

The ODE is rewritten in the standard linear form $y' + p(t)y = g(t)$:

$$y' - 2y = t^2e^{2t} \quad \Longrightarrow \quad p(t) = -2, g(t) = t^2e^{2t}.$$

For $g(t) = 0$: $y' - 2y = 0 \Rightarrow y' + (-2)y = 0$.
Using the formula $y_h = Ae^{-\int p(t)dt}$:

$$y_h = A\exp\left[-\int(-2)dt\right] = Ae^{2t}.$$

Assume

$$y = A(t)e^{-\int p(t)dt} = A(t)e^{2t}.$$

Then the method gives

$$A'(t) = g(t)\exp\left[\int p(t)dt\right]. \quad \text{(variation of parameters for 1st-order linear ODE)}$$

Compute:

$$\int p(t)dt = \int(-2)dt = -2t,$$

so

$$A'(t) = t^2e^{2t}\cdot e^{-2t} = t^2 \quad \Longrightarrow \quad A(t) = \int t^2dt = \frac{t^3}{3} + C.$$

$$\therefore y(t) = A(t)e^{2t} = \left(\frac{t^3}{3} + C\right)e^{2t}$$

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## 6

$$\frac{dy}{dx} = \frac{x^2}{1+y^2} \quad \Longrightarrow \quad (1+y^2)\frac{dy}{dx} = x^2 \quad \Longrightarrow \quad (1+y^2)dy = x^2dx$$

so the ODE is **separable**.

$$\int(1+y^2)dy = \int x^2dx \quad \Longrightarrow \quad y + \frac{y^3}{3} = \frac{x^3}{3} + C, \qquad C \in \mathbb{R}.$$

Here $f(x,y) = \dfrac{x^2}{1+y^2}$ and $\dfrac{\partial f}{\partial y} = -\dfrac{2x^2y}{(1+y^2)^2}$ are continuous everywhere, so for any initial condition $y(x_0) = y_0$, the IVP has a **unique** (at least local) solution by the existence–uniqueness theorem.

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## 7

### Case 1: $a \neq 0$

**If** $cy + d \neq 0$**:** Separating variables:

$$\frac{cy+d}{ay+b}dy = dx.$$

Integrating gives:

$$\int \frac{cy+d}{ay+b}dy = \int\left(\frac{c}{a} + \frac{ad-bc}{a}\cdot\frac{1}{ay+b}\right)dy = \frac{c}{a}y + \frac{ad-bc}{a^2}\ln|ay+b|.$$

Hence the **implicit general solution** is

$$\boxed{\frac{c}{a}y + \frac{ad-bc}{a^2}\ln|ay+b| = x + C,}$$

**Special subcase** $ad - bc = 0$**.** Then $cy + d = \frac{c}{a}(ay+b)$ and the ODE becomes $y' = \frac{ay+b}{cy+d} = \frac{a}{c}$ (constant, assuming $c \neq 0$), so

$$\boxed{y = \frac{a}{c}x + C.}$$

The **equilibrium solution** $y \equiv -\frac{b}{a}$ is included (it makes $ay + b = 0 \Rightarrow y' = 0$) provided $c(-\frac{b}{a}) + d \neq 0$.

### Case 2: $a = 0$

Then

$$\frac{dy}{dx} = \frac{b}{cy+d}.$$

#### (i) $c \neq 0$

Separating variables:

$$(cy+d)dy = bdx.$$

Integrating:

$$\frac{c}{2}y^2 + dy = bx + C,$$

so

$$\boxed{\frac{c}{2}y^2 + dy = bx + C.}$$

### (ii) $c = 0$

Then $\dfrac{dy}{dx} = \dfrac{b}{d}$ (assuming $d \neq 0$), hence

$$\boxed{y = \frac{b}{d}x + C.}$$

## Domain remark

Any solution must stay in a region where $cy + d \neq 0$ (i.e., $y \neq -d/c$ when $c \neq 0$); the differential equation is not defined on that horizontal line.

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# 8

## (a)

By Torricelli's principle, the efflux speed equals the speed a particle gains by falling freely from height $h$. For free fall (no drag), conservation of mechanical energy gives

$$mgh = \frac{1}{2}mv^2 \quad \Longrightarrow \quad v = \sqrt{2gh}.$$

## (b)

Let $V(t)$ be the volume of liquid in the tank. If the cross-sectional area at height $h$ is $A(h)$, then an infinitesimal drop $dh$ changes volume by

$$dV = A(h)dh \quad \Rightarrow \quad \frac{dV}{dt} = A(h)\frac{dh}{dt}.$$

The *volumetric* outflow rate is (actual jet area smaller than the orifice area $a$)

$$Q_{\text{out}} = (\text{area of jet}) \times (\text{speed}) = (\alpha a)v.$$

Mass/volume balance ("rate of decrease in tank volume = outflow rate") gives

$$\frac{dV}{dt} = -Q_{\text{out}} = -\alpha a v.$$

Using part (a), $v = \sqrt{2gh}$, the equation

$$A(h)\frac{dh}{dt} = -\alpha a\sqrt{2gh}$$

is obtained.

## (c)

**Given:** height above outlet $H = 3$ m, tank radius $R = 1$ m, outlet radius $r = 0.1$ m, $\alpha = 0.6$.

For a right circular cylinder, the cross-sectional area is constant:

$$A(h) = A = \pi R^2 = \pi(1)^2 = \pi.$$

The outlet area is

$$a = \pi r^2 = \pi(0.1)^2 = 0.01\pi.$$

So the ODE becomes separable:

$$A\frac{dh}{dt} = -\alpha a\sqrt{2gh} \quad \Longrightarrow \quad \frac{dh}{\sqrt{h}} = -\frac{\alpha a}{A}\sqrt{2g}\,dt.$$

Integrating from $t = 0$ (when $h(0) = H = 3$) to $t = T$ (when $h(T) = 0$):

$$\int_3^0 \frac{dh}{\sqrt{h}} = -\frac{\alpha a}{A}\sqrt{2g}\int_0^T dt.$$

Computing the left side:

$$\int \frac{dh}{\sqrt{h}} = 2\sqrt{h} \quad \Rightarrow \quad \Big[2\sqrt{h}\Big]_3^0 = 2(0 - \sqrt{3}) = -2\sqrt{3}.$$

Hence

$$-2\sqrt{3} = -\frac{\alpha a}{A}\sqrt{2g}\,T \quad \Rightarrow \quad T = \frac{2\sqrt{3}}{(\alpha a/A)\sqrt{2g}}.$$

Now $\dfrac{a}{A} = \dfrac{0.01\pi}{\pi} = 0.01$, so

$$T = \frac{2\sqrt{3}}{0.01\alpha\sqrt{2g}}.$$

Using standard gravity $g \approx 9.8\ \text{m/s}^2$:

$$T = \frac{2\sqrt{3}}{0.01(0.6)\sqrt{2 \times 9.8}} \approx 130\text{s}.$$

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## 9

Let $x(t)$ be the rocket's height above Earth's surface, so its distance from Earth's center is

$$r(t) = R + x(t), \qquad v(t) = \frac{dx}{dt}.$$

By Newton's law of gravitation, the gravitational acceleration magnitude at radius $r$ is

$$\frac{GM}{r^2}.$$

Since $g = \dfrac{GM}{R^2}$ at the surface:

$$x''(t) = \frac{dv}{dt} = -\frac{GM}{(R+x)^2} = -g\frac{R^2}{(R+x)^2}. \quad \text{(inverse-square gravity)}$$

### (a) Velocity $v$ as a function of height $x$

Using the chain rule

$$\frac{dv}{dt} = \frac{dv}{dx}\frac{dx}{dt} = v\frac{dv}{dx}.$$

The ODE becomes a separable first-order equation in $v(x)$:

$$v\frac{dv}{dx} = -g\frac{R^2}{(R+x)^2}.$$

Integrating from $x = 0$ (where $v = v_0$) to general $x$:

$$\int_{v_0}^{v} v dv = -gR^2 \int_0^x \frac{ds}{(R+s)^2}.$$

Computing each side:

$$\frac{1}{2}(v^2 - v_0^2) = -gR^2\left[-\frac{1}{R+s}\right]_0^x = gR^2\left(\frac{1}{R+x} - \frac{1}{R}\right).$$

Hence

$$v^2 = v_0^2 + 2gR^2\left(\frac{1}{R+x} - \frac{1}{R}\right).$$

Given $v_0 = \sqrt{2gR} \Rightarrow v_0^2 = 2gR$, simplifying yields:

$$v^2 = 2gR + 2gR^2\left(\frac{1}{R+x} - \frac{1}{R}\right) = 2g\frac{R^2}{R+x}.$$

Therefore (taking the upward branch $v > 0$ during ascent),

$$v(x) = \sqrt{\frac{2gR^2}{R+x}}.$$

### (b) Time to reach $x = 240{,}000$ mi, with $R = 4000$ mi

From part (a),

$$\frac{dx}{dt} = v(x) = \sqrt{\frac{2gR^2}{R+x}} \quad \Longrightarrow \quad dt = \frac{dx}{v(x)} = \sqrt{\frac{R+x}{2gR^2}}dx.$$

Thus the travel time to height $X$ is

$$t(X) = \int_0^X \sqrt{\frac{R+x}{2gR^2}}dx = \frac{1}{\sqrt{2gR}}\int_0^X \sqrt{R+x}dx.$$

Let $u = R + x$. Then $du = dx$, and when $x = 0$, $u = R$; when $x = X$, $u = R + X$:

$$t(X) = \frac{1}{\sqrt{2gR}}\int_R^{R+X} u^{1/2}du = \frac{1}{\sqrt{2gR}} \cdot \frac{2}{3}\left[(R+X)^{3/2} - R^{3/2}\right].$$

So

$$t(X) = \frac{2}{3}\frac{(R+X)^{3/2} - R^{3/2}}{\sqrt{2gR}}.$$

Using standard gravity $g_n \approx 9.8\ \text{m/s}^2$:
Converting to miles/sec $^2$ using $1\ \text{mi} = 5280\ \text{ft}$ and $1\ \text{mi} = 1609.344\ \text{m}$.
Equivalently, $g = 32.174\ \text{ft/s}^2 \approx 32.174/5280\ \text{mi/s}^2$.

With $R = 4000$ mi and $X = 240{,}000$ mi:

$$t \approx 1.8158 \times 10^5 \text{ s} \approx 50.4 \text{ hours} \approx 2.10 \text{ days} \approx \boxed{2 \text{ days } 2 \text{ hours } 26 \text{ minutes}}.$$

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## 10

$$t^2 y' = y^3 - 2ty \quad \Longrightarrow \quad y' + \frac{2}{t} y = \frac{1}{t^2} y^3.$$

This is a **Bernoulli equation** $y' + p(t)y = q(t)y^n$ with $n = 3$.

### 1) Trivial solution

$$y(t) \equiv 0$$

clearly satisfies the ODE.

### 2) Nonzero solutions ($y \neq 0$)

Dividing by $y^3$ yields:

$$y^{-3} y' + \frac{2}{t} y^{-2} = \frac{1}{t^2}.$$

The Bernoulli substitution $v = y^{1-n} = y^{-2}$ is used. Then

$$v' = \frac{d}{dt}(y^{-2}) = -2y^{-3} y' \quad \Longrightarrow \quad y^{-3} y' = -\frac{1}{2} v'.$$

Substituting into the equation:

$$-\frac{1}{2} v' + \frac{2}{t} v = \frac{1}{t^2} \quad \Longrightarrow \quad v' - \frac{4}{t} v = -\frac{2}{t^2}.$$

This is linear in $v$. The integrating factor is

$$\mu(t) = \exp\left( \int -\frac{4}{t} dt \right) = t^{-4},$$

so

$$(\mu v)' = (t^{-4} v)' = t^{-4} \left( -\frac{2}{t^2} \right) = -2t^{-6}.$$

Integrating yields:

$$t^{-4} v = \int -2t^{-6} dt = \frac{2}{5} t^{-5} + C \quad \Longrightarrow \quad v = \frac{2}{5} t^{-1} + C t^4.$$

Undoing the substitution $v = y^{-2}$:

$$y^{-2} = C t^4 + \frac{2}{5t} \quad \Longrightarrow \quad y^2 = \frac{1}{C t^4 + \frac{2}{5t}} = \frac{5t}{5C t^5 + 2}.$$

Hence, for $y \neq 0$,

$$\boxed{y(t) = \pm \sqrt{\frac{5t}{5C t^5 + 2}}}, \qquad t > 0,$$

together with the additional solution $y \equiv 0$.

### 3) Interval of definition (on $t > 0$)

Real-valued solutions require $5Ct^5 + 2 > 0$.

- If $C \geq 0$: then $5Ct^5 + 2 > 0$ for all $t > 0 \Rightarrow$ solution exists on $(0, \infty)$.
- If $C < 0$: there is a finite blow-up time

  $$t_* = \left( -\frac{2}{5C} \right)^{1/5} > 0$$

  where $5Ct_*^5 + 2 = 0$. Then the solution is defined on either $(0, t_*)$ or $(t_*, \infty)$ depending on the initial condition.

### 4) Existence & uniqueness

Writing $y' = f(t, y) = \dfrac{y^3}{t^2} - \dfrac{2y}{t}$, for $t > 0$ both $f$ and $\partial f / \partial y = \dfrac{3y^2}{t^2} - \dfrac{2}{t}$ are continuous on any rectangle in $(0, \infty) \times \mathbb{R}$. Therefore, for any IVP with $t_0 > 0$, a **unique** local solution exists (and it extends until it reaches a singularity such as $t = t_*$ above).

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## 11

$$y' = ry - ky^2 = y(r - ky), \qquad r > 0,\ k > 0.$$

## 1) Equilibrium solutions

Equilibria satisfy $y' = 0 \Rightarrow y(r - ky) = 0$, hence

$$\boxed{y(t) \equiv 0} \quad \text{and} \quad \boxed{y(t) \equiv \frac{r}{k}}.$$

## 2) Non-equilibrium solutions (separation of variables)

Assume $y \neq 0$ and $y \neq r/k$ on an interval. Then the ODE is separable:

$$\frac{dy}{y(r - ky)} = dt.$$

Using partial fractions:

$$\frac{1}{y(r - ky)} = \frac{1}{r}\frac{1}{y} + \frac{k}{r}\frac{1}{r - ky}.$$

Integrating both sides:

$$\int \left(\frac{1}{r}\frac{1}{y} + \frac{k}{r}\frac{1}{r - ky}\right) dy = \int dt$$

$$\frac{1}{r}\ln|y| - \frac{1}{r}\ln|r - ky| = t + C$$

$$\ln\left|\frac{y}{r - ky}\right| = rt + C_1.$$

Exponentiating and solving for $y$:

$$\frac{y}{r - ky} = Ce^{rt} \quad \Longrightarrow \quad y = (r - ky)Ce^{rt}$$

$$y(1 + kCe^{rt}) = rCe^{rt} \quad \Longrightarrow \quad \boxed{y(t) = \frac{rCe^{rt}}{1 + kCe^{rt}}}.$$

Equivalently, letting $A = \frac{1}{C}$,

$$\boxed{y(t) = \frac{r}{k + Ae^{-rt}}}.$$

This is the standard logistic-form solution.

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# 12

## (a) $(x \ln y + xy) + (y \ln x + xy)y' = 0,\ x > 0,\ y > 0$

Let

$$M(x, y) = x \ln y + xy, \qquad N(x, y) = y \ln x + xy.$$

The cross partial derivatives are:

$$M_y = \frac{\partial}{\partial y}(x \ln y + xy) = x \cdot \frac{1}{y} + x = \frac{x}{y} + x,$$

$$N_x = \frac{\partial}{\partial x}(y \ln x + xy) = y \cdot \frac{1}{x} + y = \frac{y}{x} + y.$$

Since $M_y \neq N_x$ in general (for $x > 0, y > 0$), the equation is **not exact**.

## (b) $\dfrac{x}{(x^2 + y^2)^{3/2}} + \dfrac{y}{(x^2 + y^2)^{3/2}}\dfrac{dy}{dx} = 0$

Define

$$M(x, y) = \frac{x}{(x^2 + y^2)^{3/2}}, \qquad N(x, y) = \frac{y}{(x^2 + y^2)^{3/2}}.$$

(These are smooth on $\mathbb{R}^2 \setminus \{(0, 0)\}$.)

### Step 1: Check exactness

$$M_y = x \cdot \frac{\partial}{\partial y}(x^2 + y^2)^{-3/2} = x\left(-\frac{3}{2}\right)(x^2 + y^2)^{-5/2}(2y) = -\frac{3xy}{(x^2 + y^2)^{5/2}},$$

$$N_x = y \cdot \frac{\partial}{\partial x}(x^2 + y^2)^{-3/2} = y\left(-\frac{3}{2}\right)(x^2 + y^2)^{-5/2}(2x) = -\frac{3xy}{(x^2 + y^2)^{5/2}}.$$

Thus $M_y = N_x$, so the ODE is **exact** on any region avoiding $(0, 0)$.

### Step 2: Find the potential function $\Psi$

The condition $\Psi_x = M$ is required. Integrating $M$ with respect to $x$ yields:

$$\Psi(x,y) = \int \frac{x}{(x^2+y^2)^{3/2}}dx = -\frac{1}{\sqrt{x^2+y^2}} + h(y).$$

Differentiating with respect to $y$:

$$\Psi_y = \frac{y}{(x^2+y^2)^{3/2}} + h'(y).$$

Setting $\Psi_y = N = \dfrac{y}{(x^2+y^2)^{3/2}}$ gives $h'(y) = 0$, so $h$ is constant.

Hence the implicit general solution is

$$\boxed{-\frac{1}{\sqrt{x^2+y^2}} = C}$$

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# 13

## (a) $(xy^2 + bx^2y) + (4x+y)x^2\frac{dy}{dx} = 0$

$$M(x,y) = xy^2 + bx^2y, \qquad N(x,y) = (4x+y)x^2 = 4x^3 + x^2y.$$

$$M_y = \frac{\partial}{\partial y}(xy^2 + bx^2y) = 2xy + bx^2,$$

$$N_x = \frac{\partial}{\partial x}(4x^3 + x^2y) = 12x^2 + 2xy.$$

Exactness requires $M_y = N_x$, hence

$$2xy + bx^2 = 12x^2 + 2xy \implies bx^2 = 12x^2 \implies \boxed{b = 12}.$$

Integrate $M$ w.r.t. $x$:

$$\Psi(x,y) = \int (xy^2 + 12x^2y)dx = \frac{1}{2}x^2y^2 + 4x^3y + h(y).$$

Differentiate w.r.t. $y$ and match $N$:

$$\Psi_y = x^2y + 4x^3 + h'(y) \overset{!}{=} 4x^3 + x^2y \;\Rightarrow\; h'(y) = 0.$$

So $h$ is constant and can be absorbed into $C$. Therefore the implicit general solution is

$$\boxed{\frac{1}{2}x^2y^2 + 4x^3y = C.}$$

## (b) $(ye^{2xy} + 5x) + bxe^{2xy}\frac{dy}{dx} = 0$

$$M(x,y) = ye^{2xy} + 5x, \qquad N(x,y) = bxe^{2xy}.$$

$$M_y = \frac{\partial}{\partial y}(ye^{2xy} + 5x) = e^{2xy} + y(2x)e^{2xy} = e^{2xy}(1+2xy),$$

$$N_x = \frac{\partial}{\partial x}(bxe^{2xy}) = b\Big(e^{2xy} + x(2y)e^{2xy}\Big) = be^{2xy}(1+2xy).$$

Exactness $M_y = N_x$ gives

$$e^{2xy}(1+2xy) = be^{2xy}(1+2xy) \implies \boxed{b = 1}.$$

Integrate $M$ w.r.t. $x$ (treat $y$ constant):

$$\int ye^{2xy}dx = \frac{1}{2}e^{2xy}, \qquad \int 5xdx = \frac{5}{2}x^2,$$

so

$$\Psi(x,y) = \frac{1}{2}e^{2xy} + \frac{5}{2}x^2 + h(y).$$

Differentiate w.r.t. $y$ and match $N = xe^{2xy}$:

$$\Psi_y = \frac{1}{2}(2x)e^{2xy} + h'(y) = xe^{2xy} + h'(y) \overset{!}{=} xe^{2xy} \Rightarrow h'(y) = 0.$$

Thus the implicit solution is

$$\boxed{\frac{1}{2}e^{2xy} + \frac{5}{2}x^2 = C} \quad\Longleftrightarrow\quad \boxed{e^{2xy} + 5x^2 = C_1.}$$

If explicit solution for $y$ is needed (for $x \neq 0$ and $C_1 - 5x^2 > 0$):

$$\boxed{y = \frac{1}{2x}\ln(C_1 - 5x^2)}.$$